

Module 03

Docker Basic Commands

DevOps end to end services & solutions

Agenda

- > Docker – Basic Commands
- > Docker run
- > Docker ps, Docker ps -a
- > Docker stop, start and restart Commands
- > Docker rm
- > Docker logs, Docker exe -it
- > Hands-On Lab-02 Mysql
- > Hands-On Lab-03 Jenkins

Basic Docker Commands



docker version

Check the installed Docker version.



docker info

Display system-wide
information.



docker help

Get help with
commands.

Container Management

- > **docker run:** Create and start a container
- > **docker ps:** List running containers
- > **docker ps -a:** List all containers
- > **docker stop:** Stop a running container
- > **docker start:** Start a stopped container
- > **docker restart:** Restart a container
- > **docker rm:** Remove a container
- > **docker logs:** View logs of a container

Let's explore these in detail.

Docker run

Runs a new container. If the image is not available locally, Docker pulls it from the registry.

```
docker run --name http_error -d -p 8990:80kennethreitz/httpbin
```

- > **-d:** Detached mode (background)
- > **-p 8990:80:** Redirect host port 8990 to container port 80
- > **--name:** Assigns a custom name

Docker ps

Lists containers that are currently **running**.

```
docker ps
```

Key Columns:

- **CONTAINER ID:** Unique identifier
- **IMAGE:** Image used
- **STATUS:** Up / Running time
- **PORTS:** Port mappings (e.g., 0.0.0.0:8990->80/tcp)

CONTAINER ID	IMAGE	STATUS	NAMES
a1b2c3d4e5	httpbin	Up 2 mins	http_error

Docker ps -a

Lists **all** containers, including those that are running, stopped, or exited.

```
docker ps -a
```

CONTAINER ID	IMAGE	STATUS	NAMES
2ffdd2cbb26c	final-project	Exited (137)	final-project
Exited (0)			

Useful for finding stopped containers to restart or remove.

Lifecycle Commands



docker stop

Gracefully stops a running

container
`docker stop`



docker start

Starts a container that was previously
stopped.

`docker start`



docker restart

Stops and then starts a container in
one step.

`docker restart`

Docker rm

Removes one or more **stopped** containers.

```
# Basic removal  
docker rm container_id  
  
# Force remove running container  
docker rm -f container_id
```

Bulk Removal:

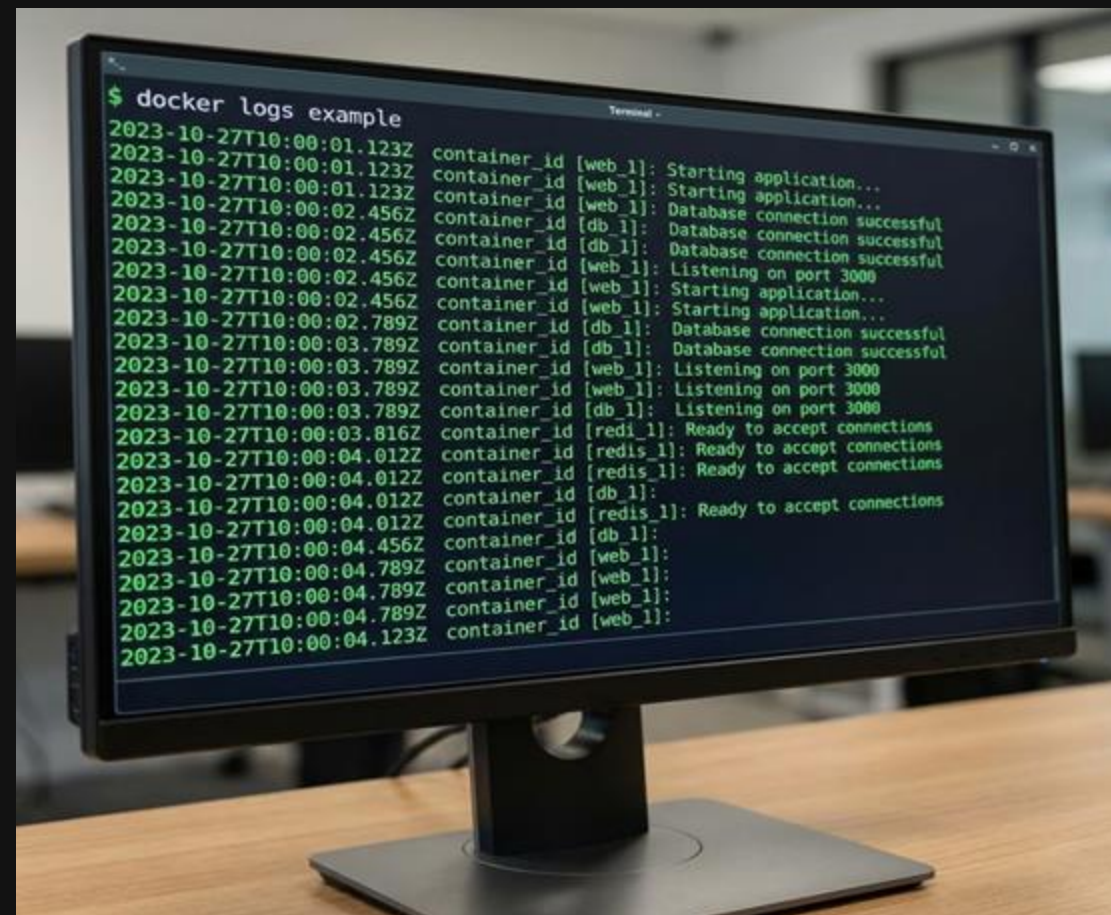
```
# Remove all stopped containers  
docker rm $(docker ps -aq)  
  
# Remove filtered containers  
docker rm $(docker ps -q -f status=exited)
```

Docker logs

Retrieves logs from a container.

- > `docker logs`
- > `--tail 100` Last 100 lines.
- > `-f` Follow log output in real-time.

```
docker logs -f --tail 100 mongo
```



Docker exec

Interactive Session

Run a command inside a running container.

```
docker exec -it mongo bash
```

Understanding Flags

- > **-i**: Interactive. Keeps STDIN open.
- > **-t**: TTY. Allocates a pseudo-terminal.

Hands-On – Lab 02: MySQL

- > Docker pull mysql
- > Docker run with port 3406
- > List of all images
- > Exec into running mysql container
- > Run SQL: show tables;



Lab 02: Cleanup



Stop

Stop MySQL container



Start

Start MySQL container



List

List all
containers



Remove

Remove
container

Hands-On – Lab 03: Jenkins

- > Run Jenkins (pull & run in one command)
- > Open Jenkins via browser
- > Create "Hello world" job
- > View Jenkins logs



Questions?

Thank you for participating.

Feel free to reach out:
<https://dinghy-e2e.com>

